



Matteo Neri

PhD candidate

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Github: <https://github.com/Mattehub/>

Summary

I am a mathematician and complex systems physicist, currently completing a PhD at the edge between **Complex Systems Physics**, **Computational Neuroscience**, and **AI**. My research focuses on how intelligent behavior emerges in both natural and artificial systems, such as the human brain, neural networks, and society. I employ graph and information theory, alongside computational modelling, to investigate learning in artificial neural networks and the human brain, and to study how distinct causal mechanisms produce higher-order information properties in complex systems. I believe a shared mathematical program on intelligence, spanning biological, social, and artificial systems, is critical for addressing AI risk and other century-defining challenges.

Education

- **MSc. in Physics of Complex Systems Physics (with honors)**, University of Turin, Italy,
Jan 2021 – Apr 2023
- **BSc. in Mathematics**, University of Bologna (UNIBO), Italy,
Thesis title: Structured Light from a Geometric Perspective (the geometry of **artificial vision**)
Nov 2017 – Oct 2020
- **Erasmus Programs**, Aix-Marseille Université (AMU), France, Sep 2021 – Jul 2022;
Universitat de Barcelona (UB), Spain, Sep 2019 – July 2020

Research Experience

- *Visiting student*, Machine Intelligence and Neural Dynamics (MIND) lab, Department of Computing, **Imperial College London**, London, UK
Aug 2025 – Present
Nov 2024 – Dec 2024

- *PhD Candidate*, **Institut de Neurosciences de la Timone**, Marseille, France, Oct 2023 – Present
- *Research Assistant*, **Institut de Neurosciences de la Timone**, Marseille, France, Apr 2023 – Nov 2023
- *Intern Student*, **Institut de Neurosciences de la Timone**, Marseille, France, Nov 2022 – Apr 2023
- *Intern Student*, **CENTAI**, Turin, Italy, Oct 2022 – Apr 2023
- *Intern Student*, **Institut de Neurosciences des Systèmes**, Marseille, France, Mar 2022 – Sep 2022

Teaching and Scientific Communication

- **Teaching Assistant**, Aix-Marseille Université, Marseille (France), Sep 2023 - Jun 2024
 - L3S5 - HPS5U07 - Méthodes et statistiques (36h)
 - L3S5 - HPS5U07 – Initiation à la psychologie Mathématique (30h)
- **Math Tutor**, Associazione diamo i numeri (~120 hours in total), Bergamo (Italy), 2021–Present
- **The AI for the people podcast** – online, Sep 2021 - Sep 2022

I participated in the organization of a podcast. The main topics discussed relate to **AI safety** and the value of AI for the future of humanity.

Grants and Awards

- *BIFONDS travel grant* (2025) — Funding for 3 months exchange, **~€3000**
- *NeuroschooL PhD funding* (2023) — PhD scholarship, **~€100K**
- *Interdisciplinary College IK* (2023) — Full stipend recipient, **~€400**
- *Poster award at NeuroDays* (2023) — prize of **€150**

Open-source software (peer-reviewed in JOSS)

Matteo Neri, Dishie Vinchhi, Christian Ferreyra, Thomas Robiglio, Onur Ates, Marlis Ontivero-Ortega, Andrea Brovelli, Daniele Marinazzo, and Etienne Combrisson (2024) *HOI: A Python toolbox for high-performance estimation of Higher-Order Interactions from multivariate data*. **Journal of Open Source Software (JOSS)**

Article DOI: <https://doi.org/10.21105/joss.07360> Github: [brainets/hoi: Higher-Order Interactions](https://github.com/brainets/hoi)

Maxime Lucas, A Townsend-Teague, **Matteo Neri**, Simone Poetto, A Morris, B Habermann (2023) *Phasik: a Python package to identify system states in partially temporal networks*. **Journal of Open Source Software**

Article DOI: <https://doi.org/10.21105/joss.05872> Gitlab: [habermann_lab / phasik · GitLab](https://gitlab.com/habermann_lab/phasik)

List of peer-reviewed publications

ORCID: 0009-0007-0998-552X

Thomas Robiglio, **Matteo Neri**, Davide Coppes, Cosimo Agostinelli, Federico Battiston, Maxime Lucas, Giovanni Petri (2024) *Synergistic signatures of group mechanisms in higher-order systems* **Physical Review Letters**

Article DOI: <https://doi.org/10.1103/PhysRevLett.134.137401>

Etienne Combrisson, Ruggero Basanisi, **Matteo Neri**, Guillaume Auzias, Giovanni Petri, Daniele Marinazzo, Stefano Panzeri, Andrea Brovelli (2024) *Higher-order and distributed synergistic functional interactions encode information gain in goal-directed learning*. **Nature Communications**

Preprint DOI: <https://doi.org/10.1101/2024.09.23.614484>

Matteo Neri, Andrea Brovelli, Samy Castro, Fausto Fraiso, Marilyn Gatica, Ruben Herzog, Pedro A.M. Mediano, Ivan Mindlin, Giovanni Petri, Daniel Bor, Fernando E. Rosas, Antonella Tramacere, and Mar Estarellas (2024) *A taxonomy of neuroscientific strategies based on interaction orders*. **European Journal of Neuroscience**

Article DOI: <https://doi.org/10.1111/ejn.16676>

Claudio Runfola, **Matteo Neri**, Daniele Schön, Benjamin Morillon, Agnès Trébuchon, Giovanni Rabuffo, Pierpaolo Sorrentino, Viktor Jirsa (2024) *Complexity in speech and music listening via neural manifold flows* **Network Neuroscience**

Article DOI: https://doi.org/10.1162/netn_a_00422

Hasnae Agouram, **Matteo Neri**, Marianna Angiolelli, Damien Depannemaecker, Jyotika Bahuguna, Antoine Schwey, Jean Régis, Romain Carron, Alexandre Eusebio, Nicole Malfait, Emmanuel Daucé, and Pierpaolo Sorrentino (2024) *L-Dopa induced changes in aperiodic bursts dynamics relate to individual clinical improvement in Parkinson's disease* **npj Parkinson's Disease**

Preprint DOI: <https://doi.org/10.1101/2024.06.14.24308683>

Carlos Coronel-Oliveros, Marilyn Gatica, Rubén Herzog, **Matteo Neri** (2025) *An integrated computational framework for diversity-sensitive personalized medicine* **IBRO Neuroscience**

Preprint DOI: <https://doi.org/10.1016/j.neuroscience.2025.08.019>

Review & Resubmit (R&R) major or minor revision in process

Andrea Santoro, **Matteo Neri**, Simone Poetto, Davide Orsenigo, Matteo Diano, Marilyn Gatica, Giovanni Petri (2025) *Beyond Pairwise Interactions: Charting Higher-Order Models of Brain Function* **R&R Nature Communications**

Preprint DOI: <https://doi.org/10.1101/2025.06.24.661306>

Matteo Neri, Claudio Runfola, Noemie AG te Rietmolen, Pierpaolo Sorrentino, Daniele Schon, Benjamin Morillon, Giovanni Rabuffo (2023) *Neuronal Avalanches in Naturalistic Speech and Music Listening* **R&R Imaging Neuroscience**

Preprint DOI: <https://doi.org/10.1101/2023.12.15.571888>

Camille Mazzara, Giuditta Gambino, Gian Marco Duma, **Matteo Neri**, Giuseppe Giglia, Tommaso Piccoli, Fabrizio Guajana, Albert Comelli, Antonino Tuttolomondo, Michele Migliore, Pierpaolo Sorrentino (2025) *Brain Fluidity as a Functional Marker of Tau-Related Neurodegeneration in Alzheimer's Disease*

Preprint DOI: <https://doi.org/10.1101/2025.07.04.25330744>

Yuri Gardinazzi, Roger González March, Suprabhath Kalahasti, Andrea Montaña Ramirez, **Matteo Neri**, Cicely Nguyen, Giovanni Palermo, Erik Weis, Katharina Ledebur, Elma Dervić (2025) *Characterization of diseases in temporal comorbidity networks*

Preprint DOI: <https://arxiv.org/html/2506.22136v1>

Talks and presentations

- **Tutorial for Beyond the Edge Workshop**
Marseille, France (October 2025)
I presented a tutorial (30min) about our toolbox HOI (<https://brainets.github.io/hoi/>).
- **Conference on Complex Systems**
Siena, Italy (September 2025)
The biggest conference on complex systems.
My contribution: **Regular talk and satellite talk**
- **Complexity72h**
Madrid, Spain (June 2025)
I presented a tutorial (30min) about our recently published Python toolbox (<https://brainets.github.io/hoi/>).
- **HOI-BrainMod**
Institute for Advanced Studies (IAS), Amsterdam, Netherlands (May 2025)
Workshop on Brain Modeling and Higher-Order Interactions
(<https://ias.uva.nl/content/events/2025/05/hoi-brainmod-workshop-on-brain-modeling-and-high-order-interactions.html>).
My contribution: **Organizer, talk, and a tutorial.**
- **Interdisciplinary College (IK) 2024**
Interdisciplinary forum that combines education and research in fields like AI, neuroscience, and cognitive science.
My contribution: **Talk and tutorial** (~1.5 hours)
- **International School and Conference on Network Science (NetSciX 2024)**
A major international conference focusing on network science across disciplines such as physics, biology, and social systems.
My contribution: **Regular talk**
- **International School and Conference on Network Science (NetSci 2023)**
The flagship annual event for network scientists worldwide.
My contribution: **Regular talk.**

Poster Presentations

- **Statistical Physics of Cognition**
Institute of Physics, London (November 2024)
This event brings together physicists and cognitive scientists to explore the role of statistical physics in understanding cognition. My contribution: poster presentation.
- **BrainModes 2024**
Bilbao, Spain (November 2024)
A highly specialized conference focused on brain dynamics and modes of neural activity. My contribution: poster presentation.
- **Association for the Scientific Study of Consciousness (ASSC27, 2024)**
Tokyo, Japan (2024)
This leading conference focuses on advancing the scientific understanding of consciousness. Contributed by creating a poster, presented on-site by co-authors.
- **Organization for Human Brain Mapping (OHBM 2024)**
Seoul, South Korea (2024)
Designed a poster that was presented on-site by co-authors. [Abstract link](#)
- **Computational and Systems Neuroscience (Cosyne 2024)**
Lisbon, Portugal (February 2024)
A premier conference for computational and systems neuroscientists, known for its high selectivity and impact. Presented a poster.
- **NeuroDays 2023**
Marseille, France (November 2023)
An interdisciplinary neuroscience symposium showcasing regional and international talent. Presented a poster, awarded **2nd best poster**.
- **Interdisciplinary College (IK) 2023**
Marseille, France (November 2023)
A unique conference blending neuroscience, psychology, and computer science, emphasizing interdisciplinary education and collaboration. Presented a poster.

Extracurricular

- **CAM – Alba (CN), Italy – Sep 2015 - Jun 2017.**

I conducted didactic activities as a volunteer.
- **Bassist and rapper – 2012 - present**

I started to play classical guitar in 2008. A few years later, I created with some friends a jazz band, in which I play the **bass** and **sing**. Recently, I also started writing and publishing **rap** songs, following the vibes of indie and jazz.